DEEP LEARNING – BASED
WEED AND CROP DETECTION IN AGRICULTURAL FIELDS

MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

Ms. Banala Naga Lakshmi (228W1A12D4)

Ms. Vangala Pavani (238W5A1221)

Under the Guidance of

Y Kalyan Chakravarti

Assistant Professor

DEPARTMENT OF INFORMATION TECHNOLOGY

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CERTIFICATE

This is to certify that this project report titled **"DEEP LEARNING – BASED WEED AND CROP DETECTION IN AGRICULTURAL FIELDS"** is a Bonafide record of work done by **Ms. Banala Naga Lakshmi (228W1A12D4), Ms. Vangala Pavani (238W5A1221)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year2025-26.

(Y Kalyan Chakravarti)	**(Dr. M. Suneetha)**
Assistant Professor	Dean R&D, Professor &Head
Dept. of Information Technology	Dept. of Information Technology

EXTERNAL EXAMINER SIGNATURE

Date of examination:

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R. SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S. No	Item	Description
1	**Project Title**	**DEEP LEARNING – BASED WEED AND CROP DETECTION IN AGRICULTURAL FIELDS**
2	**Student Names & Numbers**	**Ms. Banala Naga Lakshmi (228W1A12D4)** **Ms. Vangala Pavani (238W5A1221)**
3	**Name of The Guide**	**Y Kalyan Chakravarti (Assistant Professor)**
4	**Research Group**	**Deep learning and agriculture-based solutions**
5	**Application Area**	**Agricultural/Environment/Health care/etc..**
6	**Aim of the Project**	**To Develop an automated deep learning-based system to detect weeds in agricultural field images and estimate their density.**
7	**Project Outcomes**	**A reliable system capable of accurately detecting multiple weeds and analyzing their density across field regions.**

Student Signatures:

1.

2.

Signature of the Guide:

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TABLE OF CONTENTS

ACKNOWLEDGEMENT	III
LIST OF FIGURES	VI
LIST OF TABLES	VII
LIST OF EQUATIONS	VIII
ABSTRACT	9
CHAPTER-1 Introduction	10
1.1Origion of the Problem	10
1.2 Basic definitions and Background	12
1.3 Problem Statement with Objectives and Outcomes	13
1.4 Applications	15
CHAPTER-2 Review of Literature	16
2.1 Description of Existing Systems	16
2.2 Summary of Literature Study	18
CHAPTER-3 Proposed Method	22
3.1 Design Methodology	22
3.2 System Architecture Diagram	25
3.3 Description of Algorithms	27
3.4 Description of datasets, Requirement specification	42
CHAPTER-4 Results and Observations	45
4.1 Stepwise description of Results	45
4.2Test case results/Result Analysis	58
4.3Observations from the work	62

CHAPTER-5 Conclusion and Future work 64

5.1 Conclusion 64

5.2 Future study 65

References

Appendix

LIST OF FIGURES

Figure **Page**

Figure:3.1 UML Representation 23
Figure:3.2 Architecture Diagram 25
Figure:4.1 Performance Curve Comparison for Resnet50 Model 46
Figure:4.2 ResNet50 Model – Confusion Matrix 47
Figure:4.3 Classification Report for ResNet50 Model 47
Figure:4.4 Performance Curve Comparison for DINOv2 Model 49
Figure:4.5 DINOv2 Model – Confusion Matrix 50
Figure:4.6 Classification Report for DINOv2 Model 50
Figure:4.7 Weed and Crop Detection with Bounding Boxes 51
Figure:4.8 Outputs of Weed and Crop Detection with Confidence Scores – 1 52
Figure:4.9 Outputs of Weed and Crop Detection with Confidence Scores – 2 53
Figure:4.10 YOLOv8 Confusion Matrix 54
Figure:4.11 Faster R-CNN Confusion Matrix 55
Figure:4.12 Training and Validation Results 56

LIST OF TABLES

Table Name	Page
Table :2.1 Summary of Literature Review	18

LIST OF EQUATIONS

Equation	Brief Description	Page
Eq-1	Weed Density	56
Eq-2	Accuracy	59
Eq-3	Precision	59
Eq-4	Recall	59
Eq-5	F1-Score	59

ABSTRACT

Weed problem is especially significant, considering that such plants have a negative impact on agricultural production, consuming nutrients from crops. Contemporary solutions related to weed detection are primarily based on visual inspection of the area, which consumes many resources, time, and energy and produces errors when dealing with large areas of farmland. Thus, the creation of a model for weeds detection and estimation utilizing deep learning techniques is suggested in the current project.

When working on designing the suggested framework, it is crucial to implement contemporary object recognition models such as YOLOv8, Faster R-CNN, and EfficientDet to identify weeds and crops. Furthermore, it is important to annotate and train the models utilizing information obtained from the dataset with photographs of agricultural lands and evaluate their efficiency and accuracy.

One more issue to pay attention to while developing the framework for weeds detection and estimation is the opportunity to estimate the concentration of weeds in particular areas of farmland. In this way, it will be possible to recommend optimal ways to solve weed-related problems.

Thus, it seems appropriate to conclude that the current project is centered around the development of a framework for weeds detection and estimation using deep learning techniques.

Key words: Deep Learning, Weed Detection, Precision Agriculture, Object Detection, YOLOv8, Faster R-CNN, EfficientDet

CHAPTER-1

INTRODUCTION

1.1 Origin of the Problem

Agriculture is an integral aspect which guarantees food security and economic stability of a country. There is a considerable share of the population who depend on agriculture for their livelihood, thereby making it crucial to implement effective practices of agriculture. The practice of agriculture is faced with numerous challenges, which affect the productivity and management of the farm. One of the major challenges associated with modern agriculture is that of uncontrolled weeds in the fields of crops. Weeds refer to unwanted plants which tend to grow in farms and compete with crops for nutrients, water, sunlight, and space.

The traditional way in which farmers used to control weeds was by means of inspecting their fields manually and treating them with herbicides. This traditional approach has been used for many years, although it is laborious and imprecise. Inspecting the fields manually takes too much time and labor; besides, there is also a likelihood of human error while detecting weeds. There is the probability that farmers will fail to detect the nature of weed growth in their farms and subsequently administer inadequate treatment to control them.

Another problem associated with weed management is accurate weed density estimation in the different parts of the field. The term weed density implies the ratio between the number of weeds and crops in a certain part of the field. It is one of the essential factors which influence the necessity and degree of the weed elimination process. In case there are few weeds in a certain part of the field, then there is no need to do anything, while in case of high density, it becomes urgent to apply some measures. Wrong density estimation leads either to overuse of herbicides or to insufficient weed elimination. In both cases, there are negative consequences – increasing expenses for crop production and harming the environment (in the first case), and decreasing the efficiency (in the second).

Nowadays, thanks to deep learning technologies, it is possible to solve such problems quite easily. With the help of modern systems, it became possible to analyze agricultural pictures and classify crops and weeds. Such systems can learn how to differentiate between various plants based on such features as leaf form and colors.

Not only weed detection but estimation of their density is another benefit that farmers can receive with the help of such automated systems. After detecting the presence of certain quantity of weeds relative to crop in the image, the system will be able to estimate the density of weeds and classify them according to various levels, such as low, medium, and high densities. Such information will allow farmers to determine the level of weed contamination and act accordingly. Farmers can save both time and money by applying herbicides not to the whole field, but only to those areas that have been affected by weeds.

It should also be noted that automated detection and density estimation systems will help in proper resource allocation. In case farmers know how many weeds there are in a particular field and what their density is, they will be able to use their water, fertilizers, and other substances more efficiently. Thus, these techniques will contribute to increased productivity of crops and sustainable development practices in farming. Large volumes of images and data may be processed rapidly and effectively.

One more significant benefit offered by the technology is the high degree of consistency and objectivity. In contrast to manual methods based on human experience, deep learning algorithms deliver consistent results and eliminate possible subjectivity issues. Additionally, the application of artificial intelligence allows for detecting weed development at the early stage when intervention is still possible and can avoid the further aggravation of the issue.

Thus, in conclusion, it should be mentioned that the utilization of weed detection and density estimation technologies based on deep learning can provide a comprehensive approach to addressing a number of major problems related to agricultural production. First, the system can improve the accuracy and efficiency of weed detection. Second, it

does not require any additional labor force. Finally, it can help to make proper decisions regarding the further actions.

1.2 Basic definitions and Background

1. **Weed and Crop:**
   The weeds refer to the undesired plants which usually grow together with the crops in the agricultural fields. They compete for necessary resources like nutrients, water, light, and space. The competition impacts the growth of crops and thus reduces their yield. The crops refer to the plants grown by farmers for consumption and sale purposes.
2. **Object Detection:**
   The object detection refers to the process whereby several objects in an image are detected and located. The method involves classification of the objects and determination of their exact location through drawing of boundaries around them. Object detection plays a critical role in the project by detecting the exact locations of weeds and crops within field images.
3. **Deep Learning:**
   Deep learning is an advanced technology within the domain of artificial intelligence. The technology utilizes multi-layered neural networks to identify patterns from vast amounts of data. Deep learning is particularly suitable for processing of images since it enables the automatic extraction of features such as texture and shape.
4. **Weed Density Estimation:**
   The estimation of weed density involves estimating the amount of weeds that exist in a particular region within a farm field. This is used to determine how serious the weed infestation problem is and what actions need to be taken accordingly. The correct estimation of weed density will ensure that resources like weed killers are used efficiently.
5. **Precision Agriculture:**
   Precision agriculture is an innovative technique employed by farmers to enhance their agricultural production and efficiency. Weed detection and

density estimation in this project have helped farmers make informed decisions and conserve energy.

1.3 Problem Statement with Objectives and Outcomes

Problem Statement:

The invasion of weeds is among the major challenges that farmers encounter. This problem has significant effects on the development and productivity of the crops. As a result, the management of the weeds becomes essential. Weeding acts as a critical factor in protecting the crops from any possible effects caused by the external environment. However, there are several limitations of the traditional weed management techniques that make them inefficient.

For instance, there is a need to manually identify the weeds in the farm, and then after identification, there will be a process of spraying herbicides across the entire farm. Although this process of manually identifying the weeds has been adopted in the traditional technique within the farms, it has some difficulties. It may be quite challenging to locate the weeds in addition to identifying the different weed species. At times, the farmer cannot easily recognize the weed spots or their specific species. This would become quite hard since no steps will be put in place. In addition, the process of spraying herbicides across the farm may cause an excessive amount of chemicals.

The other important problem is that there is no accurate data about the density of weeds in different parts of the field. Without proper knowledge about how the weeds are concentrated in certain areas of the farm, farmers cannot take appropriate measures. As a result, resource utilization will be ineffective, leading to improper weed control.

In order to solve these problems, it is necessary to develop systems capable of detecting weeds and determining their density on the basis of photographs taken in the field. The use of such systems will help to save labor costs, increase the accuracy of weed detection, and make decisions based on relevant data.

Objectives:

- The main goal of this research is to create an effective deep learning algorithm that will allow the accurate identification and classification of weeds and crops on agriculture field photos while minimizing the reliance on visual identification.
- The focus of this research is on creating and evaluating various methods of objects' recognition with the purpose of choosing the best one applicable in practice.
- Special attention is paid to the quality of data set preparation, namely to proper preprocessing and labeling as well as feature selection and extraction which contribute to improved performance of the whole system.
- The algorithm is expected to be able to analyze specific features of plants including their shape, size, and texture to achieve high accuracy of their identification under varying conditions of their presence in a field.
- Moreover, the work includes estimating density of the weeds and making relevant recommendations for their management.

Outcomes:

- The new system is able to detect and localize weeds and crops from field pictures, minimizing manual inspection and work.
- The models provide reliable results with high accuracy, precision, and recall, thus allowing for consistently high efficiency under different conditions on the field.
- This system allows one to effectively monitor large areas of farmland using image processing, providing an option to apply this innovation to agriculture.
- The system can estimate weed density, providing essential information for better weed control decision-making.
- In addition, it provides useful advice that can save costs and minimize the use of herbicides.

1.4 Applications

1. **Precision Agriculture:**

   Precision agriculture technology utilizes sophisticated techniques to manage and monitor the crops effectively. With the application of the deep learning and image analysis techniques, the system can identify crops from weeds. It provides farmers an understanding of the field status hence aiding them to make informed decisions. The technique ensures accuracy, prevents wastage of resources, and enhances efficiency and production.

2. **Smart Weed Control:**

   With this technology, precise weed management is realized through the identification of exact locations of the weeds in various parts of the field. Therefore, the farmer will spray only the affected regions rather than spraying the whole field indiscriminately. Such approach minimizes the use of harmful chemicals in the farm which results in minimizing costs and protecting the environment.

3. **Large-Scale Farm Monitoring:**

   The application of such a project will be useful in managing vast agriculture fields. This will help in analyzing field images accurately without the human errors. It helps to save on labor and reduce time taken in the process.

4. **Agricultural Robotics and Drones:**

   Integration of the proposed system with the use of drones and robotic technologies is an easy task since they can detect weeds instantly and analyze the collected data. Consequently, there will be no need to conduct manual inspection, thus increasing the speed of work. In this way, the system contributes to the development of smart farms and automation of agriculture.

5. **Decision Support Systems:**

   It is possible to get useful information about the distribution and density of weeds using the developed system. This will allow farmers to make informed decisions concerning weed management and prevent any additional costs and efforts related to this problem. Therefore, there will be effective farm management.

CHAPTER-2

REVIEW OF LITERATURE

2.1 Description of Existing Systems

[1] Paper Title: Weed Detection Using Image Processing and Machine Learning (2023)

The methodology used in this research paper is that of an ordinary or traditional methodology in image processing and machine learning algorithms. The first step in implementing this algorithm involves the use of segmentation and feature extraction processes. This process identifies some specific characteristics such as shape and texture of the weeds. Moreover, the classification algorithm will involve the use of different classifiers such as Support Vector Machine (SVM) and Random Forests in order to distinguish between weeds and crops. Various experiments were carried out using field data set of agricultural fields with respect to certain criteria. This algorithm gives reasonable results under controlled conditions; however, its performance is poor in practical fields due to lighting and overlap problems.

[2] A Survey of Deep Learning Techniques for Weed Detection from Images (2022)

The paper summarizes several techniques based on deep learning algorithms designed to detect weeds, which are classified into three categories: classification, detection, and segmentation. The paper also evaluates different neural network architectures and studies their performance using various agriculture datasets. It demonstrates the advantages of utilizing deep learning models compared to machine learning algorithms in dealing with complicated image patterns. The paper further explains the advantages of transfer learning algorithms in terms of their ability to achieve high accuracy despite limited training samples. The paper addresses some of the problems in this field, such as class imbalance, environmental changes, and real-time deployment.

[3] Deep Learning Techniques for Weed Detection in Agricultural Environments (2023)

The present paper reviews various deep learning techniques for detecting weeds under

actual agricultural conditions. The study examines both detection and classification based approaches and uses field images in their comparison. The focus of this paper is on assessing the efficiency of the proposed methods with regard to the accuracy, scalability, and environmental adaptability of these approaches. Detection-based approaches are considered to be better choices for practical use because they are able to detect weeds more accurately. It is also emphasized that experiments should be carried out under realistic conditions, not artificial datasets. Transfer learning and efficient training procedures are also discussed.

[4] Precision Agriculture with Weed Detection Using Deep Learning (2024)

This study deals with the usage of deep learning algorithms for the purpose of precision agriculture. In this context, the research utilizes classification models based on transfer learning that have been trained using agricultural datasets. Transfer learning allows optimizing model performance while saving time needed for training. In addition, various data preprocessing and augmentation methods have been implemented to address possible changes in field conditions. Moreover, several models have been used for the purpose of evaluating their performance when detecting weeds. Thus, the presented findings reveal that transfer learning improves the efficiency of deep learning models for weed detection tasks.

[5] Deep Learning-Based Weed Detection Using Object Detection Models (2021)

The research presented in this paper presents a weed recognition method using the approach of object detection. Researchers make use of data sets with annotations to develop models which are able to detect and localize weed regions within an image. This research analyzes various detection methods on the basis of their accuracy and speed. The conclusion drawn from the findings is that detection models are accurate in the localization of weed regions within images and have proved to be useful in targeting treatments. The paper reveals that some of the models work fast whereas others are more precise.

[6] Weed Detection in Crops Using Convolutional Neural Networks (2023)

This paper proposes an approach based on a CNN method for weed detection within

farms. Preprocessing methods like resizing and normalization are used on the dataset to increase the performance of the model. Augmentation is done for coping with changes in light conditions and backgrounds. The classification of the image by using the CNN model is based on distinguishing weeds from crops. The performance of the model is determined based on accuracy and confusion matrix. Results reveal that CNN models exhibit good results in simple cases, but there are limitations for applying these models in more complicated farm conditions with different backgrounds.

2.2 *Summary of Literature Study*

Table:2.1 summary of Literature Review

SNO	TITLE	YEAR	WORK DONE	ALGORITHMS USED	CONCLUSION
1	Weed Detection Using Image Processing and Machine Learning	2023	This project implements weed detection using preprocessing, segmentation, and feature extraction techniques. Traditional ML classifiers were trained on extracted features. The dataset included agricultural field images. Evaluation was performed using	Image Processing, SVM, Random Forest	Image processing combined with machine learning achieves moderate accuracy. Classical ML models are limited in complex field conditions. The study highlights the need for deep learning-based solutions. It supports the transition toward CNN and transformer-based approaches.

			accuracy metrics. The system aimed to reduce manual inspection efforts.		
2	A Survey of Deep Learning Techniques for Weed Detection from Images	2022	Reviewed deep learning methods for weed classification, detection, and segmentation. Compared CNN and region-based models. Analyzed datasets and evaluation metrics. Discussed real-field challenges.	CNN, ResNet, VGG, YOLO, Faster R-CNN, U-Net	Deep learning methods outperform traditional ML techniques in weed detection. Transfer learning improves performance on limited datasets. Detection and segmentation models provide better spatial analysis than classification models. However, dataset imbalance and environmental variation remain key challenges.
3	Deep Learning Techniques for Weed Detection in Agricultural Environments	2023	Evaluated CNN and Vision Transformer models for weed recognition. Compared classification and detection frameworks. Tested performance under field	ResNet50, VGG16, EfficientNet, Vision Transformers, YOLOv5	CNN and transformer-based models achieve high accuracy in weed recognition. Detection systems are more suitable for real-world deployment than classification-only models. Transfer learning enhances efficiency and reduces training time. The study highlights the need for robust models under diverse environmental conditions.

			conditions. Analyzed scalability for agriculture.		
4	Precision Agriculture with Weed Detection Using Deep Learning	2024	The paper implements transfer learning-based classification for weed detection. Pretrained CNN models were fine-tuned using agricultural datasets. Multiple architectures were compared for performance. Data preprocessing and augmentation were applied. The system aimed to support precision farming applications.	VGG19, ResNet50, InceptionV3	Transfer learning significantly enhances model accuracy. Pretrained CNN architectures reduce training time and improve performance. Deep learning models are effective for automated weed recognition. The study validates the suitability of CNN models in agricultural monitoring systems.
5	Deep Learning-Based	2021	Developed detection framework with	YOLOv4, Faster R-CNN	Object detection models accurately localize weed regions. YOLO achieves high speed suitable for real-

	Weed Detection Using Object Detection Models		annotated bounding boxes. Implemented YOLO and Faster R-CNN. Measured detection accuracy and speed. Tested on field images.		time applications. Faster R-CNN provides slightly higher precision. Detection-based systems are more practical than classification-only models.
6	Weed Detection in Crops Using Convolutional Neural Networks	2023	Developed CNN and AlexNet models for weed detection. Preprocessed crop field images. Trained network for classification. Applied data augmentation. Evaluated model using accuracy and confusion matrix.	CNN, AlexNet	CNN and AlexNet perform well in weed detection but struggle in complex or varied field conditions. The study highlights the need for robust models that handle diverse lighting and backgrounds in agricultural fields.

CHAPTER -3
PROPOSED SYSTEM

3.1 Design Methodology (UML representation)

1.Data Collection

Image data acquisition is the initial stage of the design of the weed detection system and is highly important for successful implementation. The task is to collect agricultural field images representing different conditions encountered in real life. Good quality data allows learning the diversity of weed species, crop arrangement, and environment (lighting, soil, background).

Data Sources:

- Publicly available agricultural databases from sources such as Kaggle and other research databases.
- Images of the field taken by mobile devices or even drones, capturing changes in real time.
- Augmented data produced via rotations, flips, and scales.

Data Collection Process:

- Gather images with weeds and crops together in diverse field settings.
- Annotate images with bounding boxes that identify the weeds and the crops.
- Balance data collection so there is no bias when the machine learns.
- Make sure images are stored in standardized JPEG/PNG format.

2. Data Preprocessing

Data preprocessing ensures that raw images are transformed into the format required for model training through enhancement and standardization. Preprocessing will enable the model to learn more effective features and improve detection accuracy.

Key Preprocessing Steps:

- Resize the image to an optimal size compatible with the model.
- Normally scale the pixels in the image to improve learning performance.
- Use data augmentation techniques like rotating, flipping, brightness changes, and zooming the image.
- Check the annotated data to ensure that weeds and crops are correctly labeled.

- Divide the data set into train, validation, and test data sets.

3. Feature Engineering

For detection based on deep learning, feature extraction will be automatically carried out by the model. Patterns learned by the model from the images help in recognizing weeds and crops.

Key Features Extracted:

- **Low level features:** edges, texture and leaf pattern.
- **Mid-level features:** color patterns, plant structures and patterns.
- **High level features:** complicated interaction patterns among crops and weeds in background.
- **Spatial Features:** Position and arrangement of weeds to estimate density.
- **Embedding Vectors:** These are feature vectors for classification or clustering.

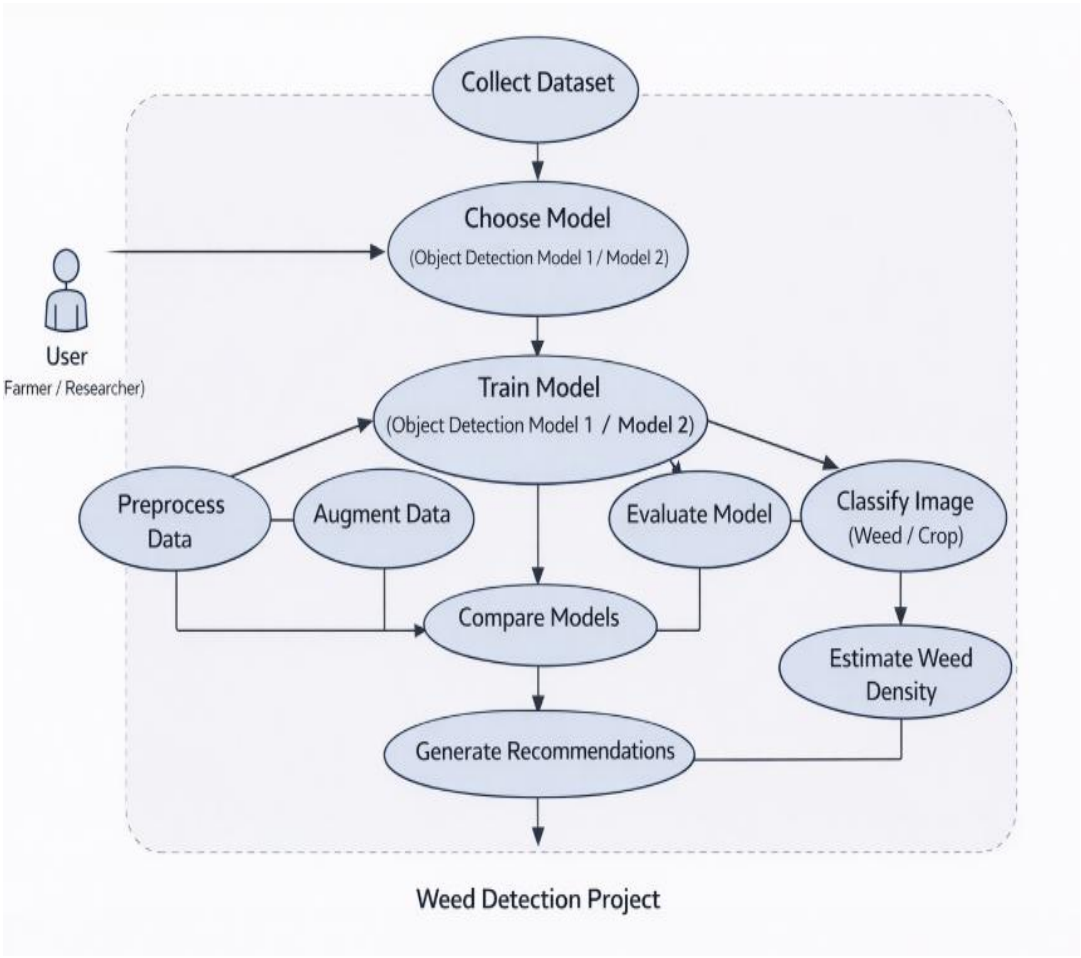

Figure:3.1 UML representation

4.Model Selection and Training

This stage involves the selection and training of object detection algorithms for the recognition and localization of weeds in field images. Several models are applied and analyzed to attain optimal results.

Models Used:

- **YOLOv8 (You Only Look Once – Version 8):** YOLOv8 (You Only Look Once – Version 8): YOLOv8 is one of the recent object detection models designed for fast performance. The model efficiently detects and identifies weeds and crops from the field image. As a matter of fact, YOLO v8 processes the whole image at once. In other words, this technique suits large areas because of its quick performance.
- **R-CNN (Region-Based Convolutional Neural Network):** This is another advanced object detection model used to detect and identify objects. The model produces region proposals based on which the classification of the weed and crop takes place. In this regard, this technique proves to be efficient in detecting weeds in field images. Although this model performs slowly compared to others, it ensures accuracy.
- **EfficientDet:** As the name suggests, Efficient Det is efficient for use in detection purposes. Indeed, this model balances between speed and performance. Therefore, the model is efficient because it utilizes optimized architecture. As such, this model will help to detect weeds and crops precisely.

Training Process:

- Train the model using labeled datasets containing bounding boxes.
- Optimize training through learning rate and batch optimization methods.
- Conduct validation while training to check progress.
- Avoid overfitting through early stopping.

5. Weed Density Estimation

Weed density estimation is the process of counting the number of weeds existing in various parts of the field. It helps in estimating the level of weed infestation.

Process:

- Count detected weed objects in images.
- Distribute them across the field.
- Classify them into low density, medium density, and high density.

6. Recommendation System

Weeds are managed using recommendations based on their population density. This assists the farmer in making informed decisions.

Recommendations Include:

- Treatment of high density areas with targeted application of herbicides.
- Lower use of chemicals in low density areas.
- Proper utilization of resources for crop management.

3.2 System Architecture Diagram

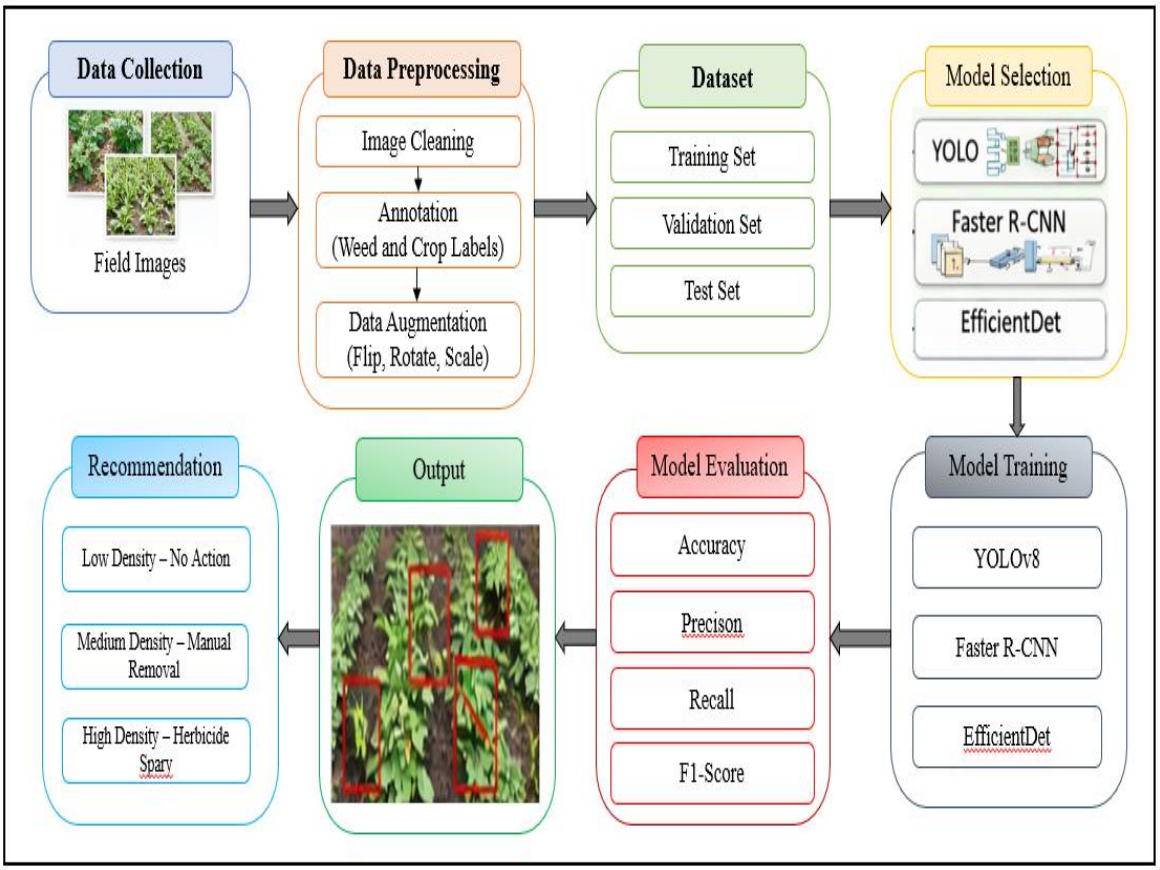

Figure:3.2 Architecture Diagram

1. Data Collection

This is a step where one collects field images that have crops and weeds in real-life situations. In some cases, images can be taken through mobile phones, drones, or agricultural images from public data sets. Such images are taken in various lighting conditions, crop growth conditions, soil, and varied weed densities. Through such variety, one can be assured that the model will become efficient enough in making generalizations based on such data. An efficient data set is a key aspect in this context.

2. Image Preprocessing

Before undergoing the training phase, all the collected data are preprocessed to enhance consistency and increase efficiency during modeling. The processes include reducing the image size into one standard size that can be used by the detection model, normalizing the pixel value, and performing image augmentation like rotation, flipping, zooming, and changing brightness levels. Furthermore, labeling the weeds and crop types is performed using a bounding box mechanism for each image.

3.Feature Extraction Using Deep Learning Models

At this stage, the DL algorithms automatically generate features that are useful for classifying the input image. Useful features generated by the algorithm include the shape, texture, and coloring of leaves and other parts of plants. They also generate complicated features like overlapping vegetation. With these features, weeds and crops can be easily identified in the field image.

4. Model Training and Validation

The pre-processed and annotated data is used to train the models. In the training process, models are trained to detect and classify weeds through error minimization during multiple training epochs. Validation data is employed to measure the performance and avoid overfitting during training. Training techniques such as changing learning rates and early stopping have been implemented for efficient training. Accuracy, precision,

recall, F1 score, and Intersection over Union are some of the evaluation metrics used in assessing the models.

5. Detection and Density Estimation

After the training process, the system will be able to identify the weeds and crops in the newly fed images through the use of bounding boxes. Weed counts will be determined based on which weed density will be predicted for the respective areas in the field. Based on the results, density classifications such as low, moderate, or high can be made.

6.Recommendation System and Output

Lastly, recommendations will be made based on the calculated weed density in the field. If the weed density is low, then nothing much needs to be done as there is very little effect caused by the weeds on the crop yield. However, in cases where there is moderate weed density, it is advisable that manual weeding should be done in order to control the weeds properly. If there is high weed density, the best recommendation would be the application of herbicides.

3.3 Description of Algorithms

3.3.1 YOLOv8 (You Only Look Once – Version 8)

- YOLOv8 is a model for real-time object detection. In this process, YOLOv8 detects and localizes objects at once.
- YOLOv8 uses the one stage object detection approach, thus giving improved speed compared to the multi-stages detection models.
- It uses anchor-free object detection approach.
- YOLOv8 model consists of three main components: backbone, neck, and detection head.
- Backbone layer is used to get essential information such as shapes and textures from the provided image.
- Neck refers to PAN-FPN model which consolidates all the multi-scale features to improve the detection of large as well as small objects.

- Detection head is used for predicting bounding boxes, objectness scores, and labels (weed or crop).
- Non-Maximum Suppression technique is used to remove redundant predictions.
- YOLOv8 has been widely recognized for its speed and accuracy.

Algorithm: YOLOv8 for Weed Detection and Density Estimation

Step 1: Initial Processing:

Pass input image x through preprocessing:

- Resize (e.g., 640x640), normalization
- Data augmentation (mosaic, flipping, scaling)
- Extract low-level features using initial convolution layers

Step 2: Backbone Feature Extraction:

Pass input through CSP-based backbone:

- Apply convolution + BatchNorm + SiLU activation
- Use C2f modules for efficient feature learning
- Extract multi-scale feature maps:

$$F = \{F1, F2, F3\}$$

Step 3: Neck (Feature Aggregation):

- Use PAN-FPN structure to fuse features
- Combine shallow and deep features
- Enhance detection of small weeds and large crops:

$$F_{agg} = PAN(F)$$

Step 4: Detection Head (Anchor-Free):

For each feature level:

- Predict bounding box:

$$B = (x, y, w, h)$$

- Predict objectness score:

$$S = P(object)$$

- Predict class probability:

$$C = P(class/object)$$

Step 6: Loss Computation:

- Classification Loss (Binary Cross Entropy)
- Localization Loss (CIoU Loss)
- Objectness Loss

Step 7: Post Processing:

- Apply Non-Maximum Suppression (NMS):

$$B_{Final} = NMS(B, S)$$

Step 9: Output:

$$\hat{y} = \{(B_i,\ C_i,\ S_i)\}$$

If $C_i = 0 \rightarrow Crop$

If $C_i = 1 \rightarrow Weed$

3.3.2 Faster R-CNN (Region-Based Convolutional Neural Network)

- Faster R-CNN is an example of two-stage object detection which boasts a great accuracy.
- The first stage involves the creation of region proposals through Region Proposal Network.
- Then, the regions are classified either as weeds or crops.
- The algorithm has a convolutional backbone that creates feature maps for the image.
- Afterwards, ROI Pooling is done to get fixed-sized feature maps out of regions.
- Classification and bounding box regressions are both accomplished at once.
- Overlapping plants in images do not pose difficulties for Faster R-CNN.
- Though it is not as fast as YOLO, the model offers more accurate results.
- This detector should be used when higher accuracy outweighs speed.

Algorithm: Faster R-CNN for Weed Detection

Step 1: Initial Processing:

Pass input image x through preprocessing:

- Resizing and normalization

- Feature extraction using backbone

Step 2: Feature Map Generation:

$$F = Backbone(x)$$

Extract spatial features such as leaf texture and shape.

Step 3: Region Proposal Network (RPN):

For each location in feature map:

- Generate anchor boxes
- Predict
    - Objectness score
    - Bounding box offsets

$$R = RPN(F)$$

Step 4: Region Proposal Filtering:

- Remove low-score proposals
- Apply Non-Maximum Suppression:

$$R_{filtered} = NMS(R)$$

Step 5 : RoI Align (Region Extraction):

- Convert proposals into fixed-size feature maps:

$$F_{roi} = RoIAlign(F,\ R_{filtered})$$

Step 6: Classification and Regression Head:

For each region:

- Classification:

$$C = Softmax(F_{roi})$$

- Bounding box refinement:

$$\mathrm{B} = Regression(F_{roi})$$

Step 7: Loss Function:

- Classification Loss (Cross Entropy)
- Localization Loss (Smooth L1)

Step 8: Output:

$$\hat{y} = \{(B_i,\ C_i)\}$$

If $C_i = 0 \rightarrow Crop$

If $C_i = 1 \rightarrow Weed$

3.3.3 EfficientDet

- EfficientDet is an efficient and scalable object detection model for balanced accuracy.
- It utilizes EfficientNet architecture for feature extraction.
- The architecture incorporates BiFPN (Bidirectional Feature Pyramid Network).
- BiFPN helps in feature fusion from various levels of features.
- Compound scaling technique is applied to change the parameters of the network.
- The detection head is used for predicting bounding boxes and class scores.
- This model has achieved a balance between accuracy and computation.
- The model works efficiently even under resource-constrained conditions.
- EfficientDet model is ideal for agricultural practices.

Algorithm: EfficientDet for Weed Detection

Step 1: Initial Processing:

- Resize input image x
- Normalize pixel values
- Apply augmentation

Step 2: Backbone (EfficientNet):

$$F = EfficientNet(x)$$

Extract multi-scale features efficiently.

Step 3: BiFPN Feature Fusion:

- Combine features in both directions (top-down & bottom-up)
- Learn importance weights:

$$F_{Fusion} = \sum w_i \cdot f_i$$

Step 4: Feature Refinement:

- Apply depthwise separable convolutions

Improve efficiency with reduced parameters

Step 5: Detection Head:

For each scale:

- Class prediction:

$$C = Sigmoid(F)$$

- Bounding box prediction:

$$B = (x, y, w, h)$$

Step 6: Loss Computation:

- Focus Loss (handles class imbalance)
- Bounding box Loss (loU / Smooth L1)

Step 7: Post Processing:

- Apply Non-Maximum Suppression (NMS):

$$B_{Final} = NMS(B)$$

Step 8: Output:

$$\hat{y} = \{(B_i,\ C_i,\ S_i)\}$$

If $C_i = 0 \rightarrow Crop$

If $C_i = 1 \rightarrow Weed$

3.3.4 Resnet50

- ResNet-50 is a 50-layers deep convolutional neural network which helps to overcome the issue of vanishing gradients.
- ResNet-50 uses residual learning, wherein each layer learns just the difference between the two mappings.
- Within a residual block there exists a skip/shortcut connection which sums the input with the output of the block.
- Skip connection helps gradient to flow smoothly, thus facilitating effective training of a very deep neural network.
- ResNet-50 consists of convolution, batch normalization and activation function.
- Network learns features hierarchically – from edges to complex features such as texture and shape.
- ResNet-50 concludes with global average pooling and dense layer.

Algorithm: Resnet50 model for Weed vs Crop Classification

Input: Training dataset D = $\{(x_i, y_i)\}$, Validation dataset D_{val}, Test dataset D_{test}, learning rate

Output: Trained Resnet50 model M.

Method:

- Resize all images to 224 * 224 and normalize pixel values.
- Initialize Resnet50 with parameters Θ; if transfer learning, load pre-trained weights.
- Replace the final layer with 1 neuron and sigmoid activation for weed and crop classification.
- Use residual blocks for forward propagation: out = x + F(x;W) .
- Freeze base layers initially.
- for epoch = 1 to N:
    - Shuffle training dataset D
    - for each mini-batch B = $\{(x_j, y_j)\}_{j=1}^{b}$
        - Compute predictions: $\hat{y}_j$ = f(x_j; Θ) using Resnet50
        - Compute loss: L = $-\frac{1}{b}\sum_{j=1}^{b}[y_j \log \hat{y}_j + (1 - y_j)\log(1 - \hat{y}_j)]$
        - Backpropagate gradients: g = ∇_{Θ}L
        - Update parameters: Θ ←Θ – ηg
    - end for
    - Evaluate validation accuracy on D_{val}
- end for
- for each image x ϵ D_{test} or new image:
    - Predict class: $\hat{y}$ = f(x; Θ) (weed or crop).
- end for
- Return trained model M(x) = f(x; Θ)

Internal Process of ResNet50 (Residual Blocks) for Weed vs Crop Classification

1. Initial Processing:

   Pass input image x through the initial convolution layer (7×7), batch normalization, ReLU activation, and max pooling to extract low-level features (edges, shapes, and textures).
2. For each residual block in ResNet50:
    a) Apply a sequence of three convolutional layers:
        - 1×1 convolution → reduces dimensions

- 3×3 convolution → extracts spatial features
- 1×1 convolution → restores dimensions
- Each convolution is followed by Batch Normalization and ReLU activation.

b) Compute the residual mapping:

F(x;W)=Conv-BN-ReLU operations on input x

c) Add skip connection (identity mapping):

$$y = F(x;W) + x$$

d) Apply ReLU activation to the summed output:

$$y = ReLU(y)$$

3. Stack residual blocks across 5 stages:
   - Stage 1: 3 residual blocks
   - Stage 2: 4 residual blocks
   - Stage 3: 6 residual blocks
   - Stage 4: 3 residual blocks
   - Each stage captures deeper and more complex features (color patterns, leaf texture, and shape differences between weeds and crops).
4. After all residual blocks:

   Apply Global Average Pooling to convert the final feature maps into a single vector representing the image features.
5. Classification Head:

   Pass the pooled feature vector through:
   - Dense layer with ReLU activation
   - Dropout for regularization
   - Final Dense layer with Sigmoid activation for binary output (weed or crop).
6. Output:

   Predicted class:

$$\hat{y} = f(x;\ \theta)$$

If $\hat{y} = 0$ → Crop

If $\hat{y} = 1$ → Weed

3.3.5 DINOv2

- DINOv2 is an unsupervised Vision Transformer (ViT) model.
- It follows a teacher-student self-distillation paradigm, wherein the two models have identical architectures.
- The model learns from the contrast between several augmentations of the same image.
- The student network attempts to replicate the features generated by the teacher model, becoming invariant to scale, color, and form.
- The weights of the teacher network are adjusted based on the exponentially smoothed weights of the student network.
- Learns sophisticated, semantics-based visual representations capable of transferring to many applications.
- Does not require any human labeling of data, thus being appropriate for use in agriculture and other sectors with low amounts of labeled data.
- Generates high-quality embeddings for classification, clustering, and segmentation applications.

Algorithm: DINOv2 model for Weed vs Crop Classification

Input: Training dataset D = $\{(x_i, y_i)\}$, Validation dataset D_{val}, Test dataset D_{test}

Output: Trained DINOv2 model M.

Method:

- Resize all images to 224 * 224 and normalize pixel values.
- Initialize DINOv2 with parameters Θ; if transfer learning, load pre-trained weights.
- Replace the final layer with 1 neuron and sigmoid activation for weed and crop classification.
- Use DINOv2 feature extractor (self-supervised backbone) for forward propagation.
- Freeze base layers initially.
- for epoch = 1 to N:
    - Shuffle training dataset D

- for each mini-batch B = $\{(x_j, y_j)\}_{j=1}^{b}$
    - Compute predictions: $\hat{y}_j$ = f(x_j; Θ) using DINOv2
    - Compute loss: L = $-\frac{1}{b}\sum_{j=1}^{b}[y_j \log \hat{y}_j + (1 - y_j)\log(1 - \hat{y}_j)]$
    - Backpropagate gradients: g = ∇_{Θ}L
    - Update parameters: Θ ←Θ – ηg
- end for
- Evaluate validation accuracy on D_{val}

- end for
- for each image x ϵ D_{test} or new image:
    - Predict class: $\hat{y}$ = f(x; Θ) (weed or crop).
- end for
- Return trained model M(x) = f(x; Θ)

For each encoder block in DINOv2

a. Compute Multi-Head Self-Attention (MHSA):
    - Derive Query (Q), Key(K), and Value(V) from patch embeddings.
    - Compute attention weights.
        - ➢ Attention(Q, K, V) = Softmax ($\frac{QK^T}{\sqrt{d_k}}$) V
    - Captures relationships between all patches globally.

b. Add & Normalize
    - Add residual (skip) connection: x’ = x + Attention(Q,K,V)
    - Apply Layer Normalization

c. Feed Forward Network (MLP):
    - Pass x’ through a two-layer MLP with GELU activation
    - Add residual connection again: y = x’ + MLP(x’)
    - Apply Layer Normalization

Feature Extraction:

- After the final encoder block, extract the [CLS] token embedding.
- This embedding represents the global images features (weed vs crop characteristics).

Output:

- The extracted feature vector F(x; W_{dino}) is passed to the classification layer.
- The classifier applies a sigmoid activation to predict:
  - 0 → Crop
  - 1 → weed

3.3.6 Random Forest

Random Forest is an ensemble algorithm that constructs multiple decision trees while training and aggregates their predictions (averaging in the case of regression problems) in order to enhance prediction performance and avoid overfitting. In your project, Random Forest is applied in order to cope with non-linear relations of features including weather, demand, supply, and price variations.

- Bagging (Bootstrap Aggregation): Different random samples are selected in order to train every tree.
- A separate tree provides a prediction, and the output will be the averaged prediction.
- The model performs very well in identifying complex relations, dealing with missing data, and minimizing variability.
- The algorithm is suitable for multivariate crop price prediction when several related factors affect crops' prices.

Algorithm: Generate a Random Forest Regressor for Crop Price Prediction

Input:

- Training Data D: A dataset of features (e.g., crop type, rainfall, supply, demand, region) and corresponding crop prices.
- n_trees: Number of decision trees to be generated.
- m_features: Number of features to consider at each split.

Output:

- A trained Random Forest regression model.

Method:

1. Initialize the forest F as an empty set.

2. For i = 1 to n_trees:

a. Draw a bootstrap sample D_i from training data D (random sampling with replacement).

b. Build a decision tree T_i using D_i:

At each node, randomly select m_features from the full feature set.

Use the best split from the selected features (based on MSE).

Continue recursively until stopping criteria are met (e.g., max depth or minimum samples).

c. Add the trained tree T_i to forest F.

3. To make a prediction for a new instance x:

a. Let each tree T_i in F predict the output y_i.

b. Return the final prediction as the average of all y_i.

4. Return the trained Random Forest model.

3.3.7 Decision Tree

Decision Tree is a supervised machine learning model that recursively partitions data depending on its features to build a tree structure. It is easily understood by users, so it can be used at the early stages for prediction or to explain why the price was predicted that way.

- It applies features like crop type, rainfall, and month to make “if-else” decisions.
- Every internal node in this tree structure is responsible for testing an attribute, branches – showing results, and leaf nodes giving a prediction of the final price.
- This method tends to overfit data on its own but can be helpful when creating

more complex models like Random Forest.

- It provides immediate understanding of what is happening.

Algorithm: Generate a decision tree from the training tuples of data partition, D

Input:

- Data partition, D, which is a set of training tuples and their associated class labels;
- Attribute_list, the set of candidate attributes;
- Attribute_selection_method, a procedure to determine the splitting criterion that "best" partitions the data tuples into individual classes. This criterion consists of a splitting_attribute and, possibly, either a split-point or splitting subset.

Output:

- A decision tree.

Method:

1. create a node N;

2. if tuples in D are all of the same class, C, then

    return N as a leaf node labeled with the class C;

3. if attribute_list is empty then

    return N as a leaf node labeled with the majority class in D; // majority voting

4. apply Attribute_selection_method(D, attribute_list) to find the "best" splitting_criterion;

5. label node N with splitting_criterion;

6. if splitting_attribute is discrete-valued and multiway splits allowed then // not restricted to

binary trees

attribute_list ← attribute_list − splitting_attribute; // remove splitting attribute

7. for each outcome j of splitting_criterion

// partition the tuples and grow subtrees for each partition

a. let D_i be the set of data tuples in D satisfying outcome j; // a partition

b. if D_i is empty then

attach a leaf labeled with the majority class in D to node N;

c. else attach the node returned by Generate_decision_tree(D_i, attribute_list) to node N;

8. endfor

9. return N;

3.3.8 LSTM (Long Short-Term Memory)

LSTM stands for Long Short-Term Memory which is a type of Recurrent Neural Network (RNN) used specifically to predict future values from a series of time-related data points. It has the ability to learn about temporal patterns and dependencies between past and future occurrences.

- The LSTM model consists of memory units having three types of gates i.e. the input gate, output gate, and the forget gate.
- Its unique ability to learn about past events and their impact on price, such as weather or season.
- Used to predict prices based on monthly data points.
- Although it takes a large dataset and more time to train but gives very precise results.

Algorithm: Generate a LSTM from the training tuples

Input:

- X_train: Input time-series features (e.g., previous prices, weather,

supply-demand sequences).

- y_train: Actual crop prices corresponding to each time step.
- epochs: Number of training iterations.
- batch_size: Size of each training batch.
- lstm_units: Number of LSTM cells in the layer.

Output:

A trained LSTM model for predicting future crop prices.

Method:

1. Normalize the input features in X_train for stable training.

2. Reshape X_train into 3D format [samples, time_steps, features].

3. Initialize an LSTM model with:

    a. One or more LSTM layers with lstm_units.

    b. Optionally, add dropout layers to prevent overfitting.

    c. A dense output layer to produce the final price prediction.

4. Compile the model with:

    a. Loss function = Mean Squared Error (MSE).

    b. Optimizer = Adam (or RMSprop).

5. Train the model using X_train and y_train for the specified number of epochs and batch size.

6. After training, use the model to predict prices by feeding future sequences.

7. Return the trained LSTM model.

3.4 Description of datasets and Tools

Description of Datasets

The dataset used in the current study is the multilevel agricultural weeds and crops dataset meant for detecting and analyzing plants in their environment. The dataset is

made up of 5,858 images that have weed and crop samples collected from various sources such as agriculture repositories, field photography, and agricultural research. The agricultural images are made up of diverse farm environments, making the model more competent when applied to the natural environment.

Each image in the data contains the objects classified by their classes, which in this case are the weeds and crops through annotated bounding boxes. The model uses this information not only to detect the plants but also locate them in the images. The agricultural images contain diverse features of leaf shapes, colors, textures, and even growth stages of the plants from the seedling stage to maturity.

In addition, the dataset takes into account the different backgrounds of the plants, soil, and light changes, and some images might have multispectral reflectance characteristics. The above features are important in providing a variety of plant images, making the models able to differentiate crop plants from weeds and estimate the number of plants.

Description of Tools and Requirements

Python: Python is the core programming language that will be used to develop this project. Python is widely known to provide flexibility and ease of use during coding while providing sufficient options for working with image data, data preprocessing techniques, and machine learning models. Therefore, it is the right choice for building a robust framework for developing the weed detection model and estimating their density levels.

PyTorch: The PyTorch framework will be the core component used to design and train the object detection models for detecting weeds in different environments. This is because the PyTorch framework comes with dynamic computational graphs, which provide sufficient flexibility to test new models without the need to change the entire code structure. In addition, it has robust GPU support, which helps to train models faster. PyTorch also comes with pretrained models that make image detection easier.

NumPy: Another tool used in developing the weed detection system is the NumPy package. This package will be utilized to perform numerical calculations needed for

different preprocessing tasks. This package works well with multi-dimensional data that can be used to represent images. This makes it the right choice for building and evaluating weed detection algorithms.

Pandas: Pandas is the library that is used in this assignment to store all relevant data sets' information like annotations, labels, and metadata. The library allows for efficient reading and manipulation of structured data in particular from CSV containing information about the bounding boxes. In turn, by providing an effective and neat way of working with the data, Pandas helps ensure that the consistency of data is guaranteed.

OpenCV: The OpenCV library is responsible for performing all kinds of preprocessing related to image manipulation. For example, OpenCV can resize images, perform normalization, and even add random noise for augmentation purposes. In addition, with the help of the OpenCV library, it is possible to visualize images, where the results of detection (with bounding boxes) are shown. Thus, using OpenCV guarantees the correct preparation of input images for the models.

Matplotlib: The Matplotlib library is helpful in visualizing the performance of the models in terms of training and evaluation. In other words, Matplotlib allows generating loss curves, accuracy statistics, and even visualized images. The use of Matplotlib will enable to analyze and interpret the results obtained by means of machine learning techniques.

CHAPTER-4
RESULTS AND OBSERVATIONS

4.1 Stepwise description of Results

Resnet50 Model:

- X-Axis: The x-axis denotes the training epoch number ranging from 1 to 25, and the plot depicts how the model learns and improves over the several rounds of training.
- Y-Axis (Loss Plot): It denotes the loss experienced by the model in terms of predicting, where low values represent that the model is learning efficiently and reducing its prediction error rate.
- Y-Axis (Accuracy Plot): It represents how accurately the model is trained and validated, indicating its efficiency in predicting the correct class of an image.
- Blue Line: Blue colored line denotes the performance of the model on the training data set, where there is an observation of gradually improving learning behavior and decreasing errors.
- Orange Line: Orange colored line denotes the performance of the model on validation data set; when orange and blue lines overlap, it indicates good learning.

Observations:

- The loss plot for both training and validation is clearly decreasing, which is evidence that the learning process is effective and results in reduction of error in all epochs.
- The loss of validation settles(Figure 4.1) at the level of 0.32-0.34, which is not far from the loss plot of training, thus implying that there is no problem of overfitting here.
- The accuracy of training reaches almost 93% while increasing from about 60%, while the accuracy of validation goes up to about 91%.
- The model is able to learn fast during initial training as indicated by the dramatic decrease in loss value and increase in accuracy rate.

- The minor deviations in accuracy in validation during some mid-epochs is nothing but natural variation in unseen validation samples.
- High and consistent accuracy alongside low loss in the final epoch demonstrate the convergence of the model.
- Well-matched training and validation curves suggest the absence of any overfitting or underfitting issues in the learning process.
- From confusion matrix results, it is clear that high percentages of correct classifications for the weed and crop categories are achieved; moreover, the presence of strong diagonals indicates correct classification.
- There is a lack of false positives (weeds classified as crops) and false negatives (crops classified as weeds), thus indicating the high reliability of the model.
- Precision and recall metrics, which are calculated on the basis of the confusion matrix, provide a proof for reliable operation of the model.

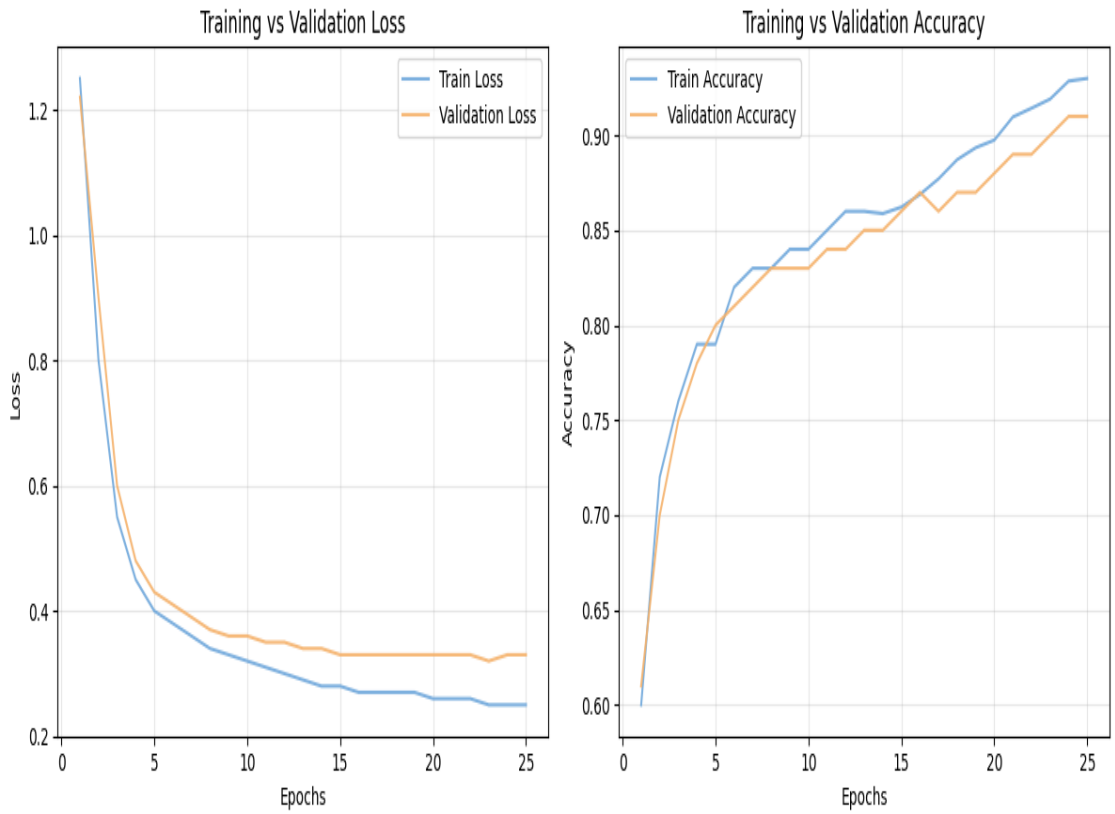

Figure:4.1 Performance Curve Comparison for Resnet50 Model

```
Classification Report:

              precision    recall  f1-score   support

       crops       0.89      0.92      0.91       220
       weeds       0.93      0.90      0.91       249

    accuracy                           0.91       469
   macro avg       0.91      0.91      0.91       469
weighted avg       0.91      0.91      0.91       469
```

Figure:4.2 Classification Report for ResNet50 Model

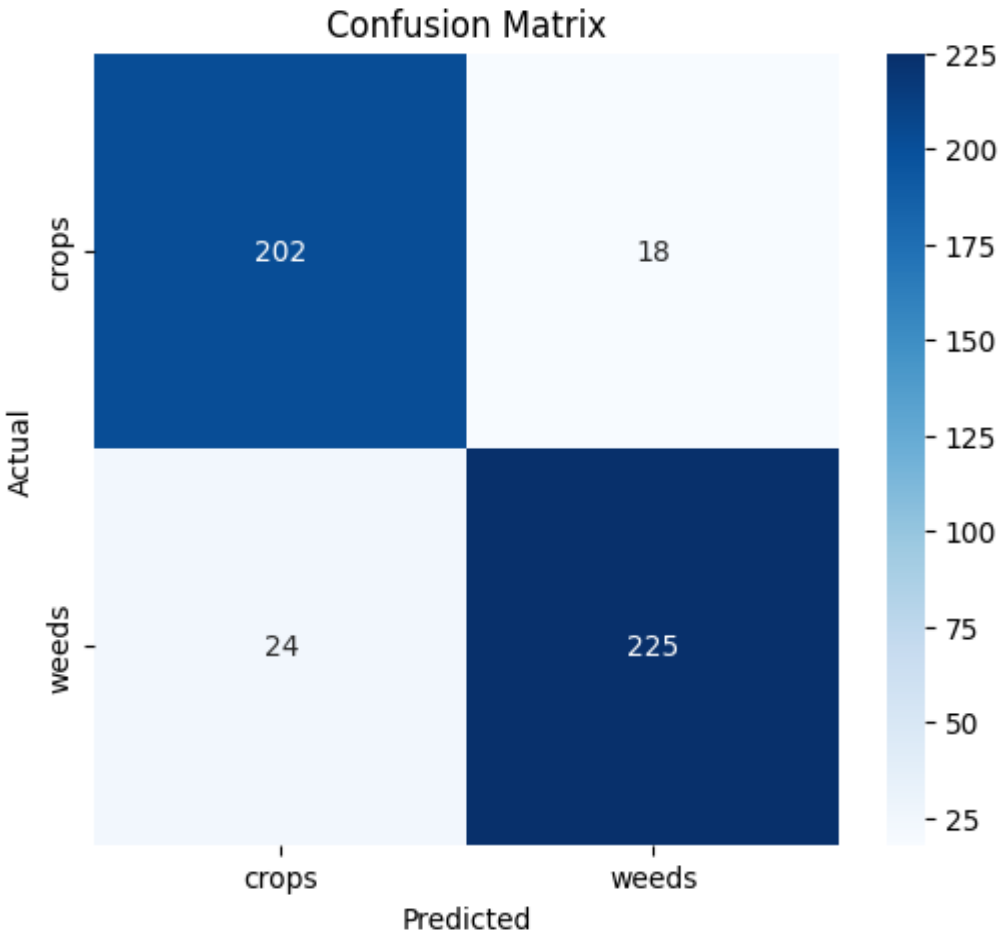

Figure:4.3 ResNet50 Model – Weed and Crop Classification Confusion Matrix

DINOv2 Model:

- X Axis: The x-axis indicates the number of training epochs (1-30), showing the gradual development of the DINOv2 model with the increasing epochs.
- Y-Axis (Loss Graph): The Y-axis in the graph above is used to show the training and validation loss, which refers to the ability of the model to perform a task effectively based on its ability to learn.
- Y-Axis (Accuracy Graph): Accuracy refers to the ability of the model to classify images correctly.
- Blue Line (Training): The blue line shows how the DINOv2 model learned from the data during each training session, as seen in a decrease in loss and an increase in accuracy.
- Orange Line (Validation): The orange line refers to how the model is able to classify images during the validation process, where a closer line is indicative of effective learning.

Observations:

- Below, both training loss curve and validation loss curve can be seen to experience gradual reductions, declining from about 1.25–1.30 in the first epoch to 0.28–0.33 in latter epochs.
- Validation loss tends to stabilize at about 0.30 to 0.34, keeping itself close to that of training loss for all training time.
- Training accuracy goes up considerably from about 60% to 89%, and validation accuracy rises from 58% to 85% within the 30 epochs.
- The model is seen to have a remarkably high learning rate during the initial 5–8 epochs, with losses dropping sharply and accuracy increasing considerably.
- Small-scale fluctuations can be observed around the middle epochs in terms of validation accuracies, but still remain within a narrow range.
- This is normal and caused by discrepancies among unseen validation data images.
- In the last training epochs, one can observe that there is high and stable accuracy (approximate to 89%) and low values of loss (for training – about 0.33; for validation – about 0.46).

- These findings indicate effective learning and good convergence on the part of the machine-learning model.
- The similarity in training and validation curve shows that there is no sign of under-fitting or over-fitting.
- It becomes clear that DINOv2 shows good performance in terms of being able to learn features from training data and generalize knowledge when dealing with unseen images.
- This statement can be supported by the observation that the confusion matrix shows strong diagonal dominance, meaning that both the crop and weed classes get classified properly most of the time.
- The number of false positives and false negatives is very low, suggesting that the machine-learning algorithm can successfully differentiate between similar plant species.
- The obtained high precision/recall scores prove that performance is consistent in terms of both classes.

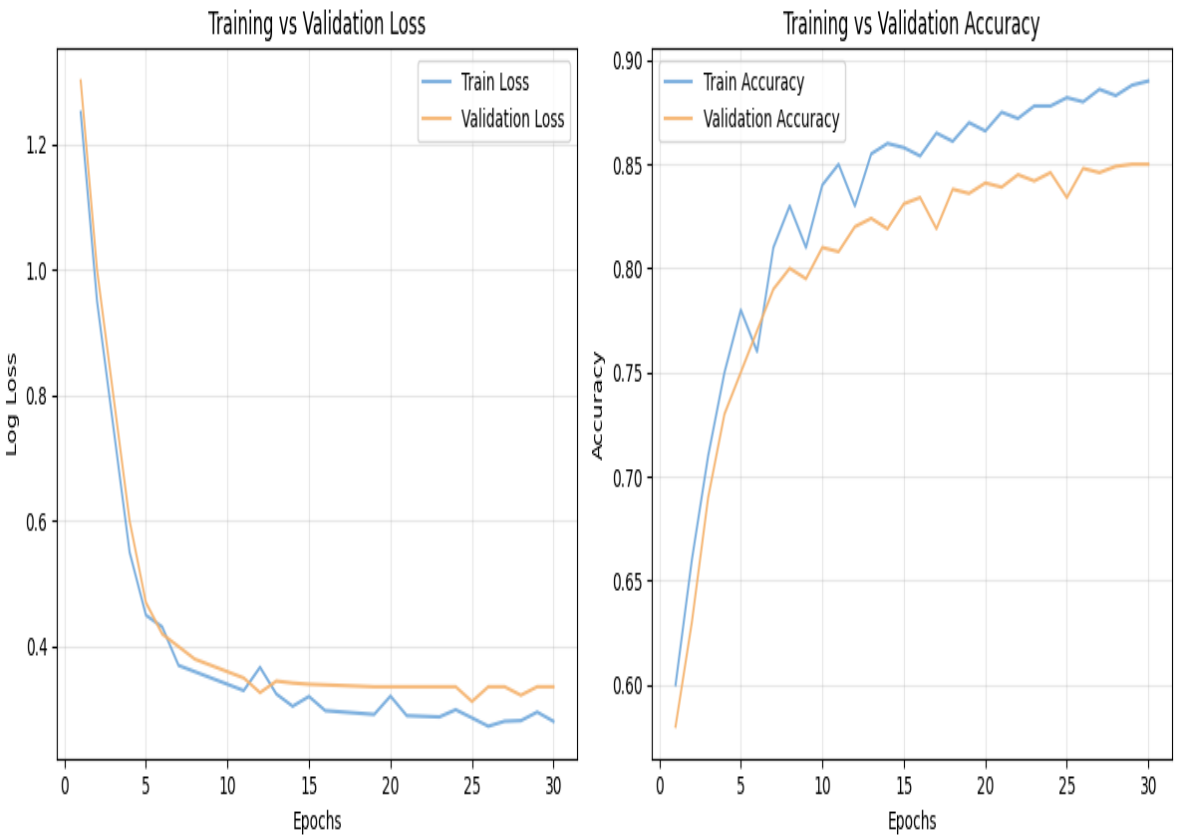

Figure:4.4 Performance Curve Comparison for DINOv2 Model

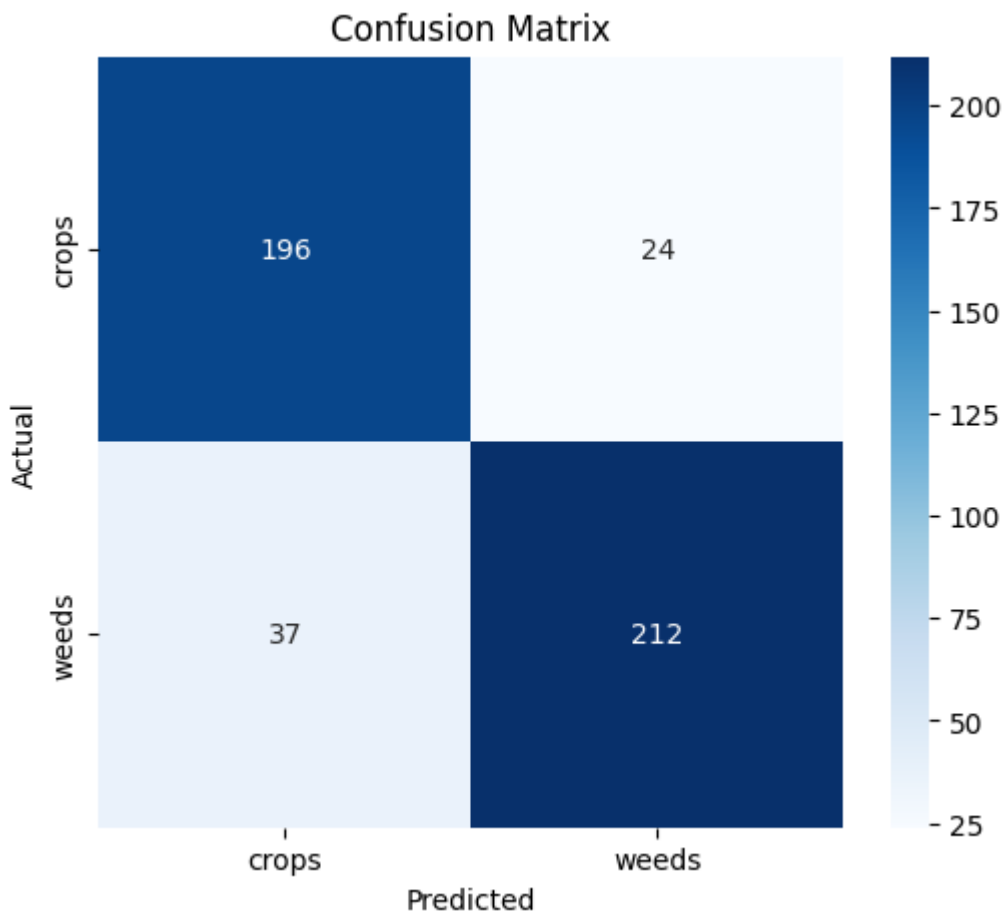

Figure:4.5 DINOv2 Model – Weed and Crop Classification Confusion Matrix

Classification Report:

	precision	recall	f1-score	support
crops	0.84	0.89	0.87	220
weeds	0.90	0.85	0.87	249
accuracy			0.87	469
macro avg	0.87	0.87	0.87	469
weighted avg	0.87	0.87	0.87	469

Figure:4.6 Classification Report for DINOv2 Model

Detection Model Performance (Training Phase):

- X-axis: The X-axis is associated with the number of training epochs (1-50). It implies that there is a series of iterations, in which the neural network improves its performance through learning some image characteristics.
- Y-axis (Loss Curves): Loss curves present bounding box, classification, and objectness losses. Their overall score indicates how the network performs in terms of detecting and classifying certain objects. Hence, the lower these loss scores, the better the model performs in predicting the output.
- Y-axis (mAP Score): The mAP score reveals the performance level of the model in detecting and classifying objects. In this case, the higher the score is, the better is the result.
- Training Curve: The training curve is connected with the performance of the model on the training set. Training curve demonstrates how the neural network learns to detect and classify objects with the use of annotated data and bounding boxes.
- Validation Curve: The validation curve presents the performance of the model on the validation set. It helps evaluate how well the network is trained since it shows whether the model can successfully process new images. Therefore, the validation curve should look like the training one.

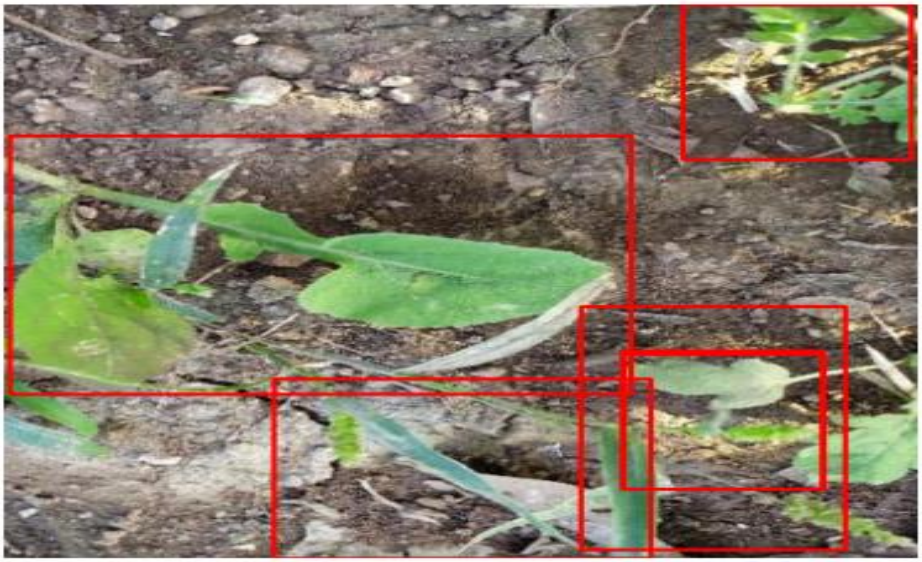

Figure:4.7 Weed and Crop Detection with Bounding Boxes

Observations:

There is a significant improvement in the performance of the model as the training proceeds, as depicted by the results above. This can be seen by the gradual decline in the value of the loss curves as the number of iterations increase. This shows that the model is learning how to correctly detect and classify the objects with minimal errors. On the other hand, there is an increase in the value of mAP score showing that there is an improvement in the model in detecting and classifying the images. The initial epochs see an almost instant improvement in the model as it learns simple features such as the shapes and sizes of the crops and weeds. Furthermore, it can be seen that the validation curve follows very closely the training curve showing that there is no problem with overfitting of the model.

Object Detection Output Analysis (Bounding Boxes):

These outputs have a clear depiction of the detection results through bounding boxes that contain:

- Classification result (Crop / Weed)
- Confidence level (e.g. 0.8, 0.9)
- The model is able to detect more than one object in an image, making it feasible for real-world scenarios.

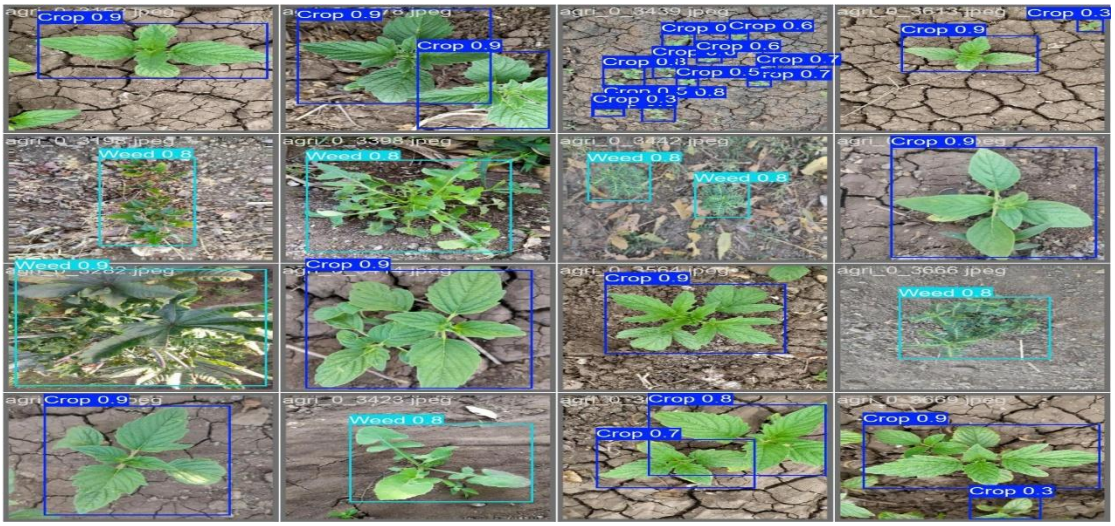

Figure:4.8 Outputs of Weed and Crop Detection with Confidence Scores - 1

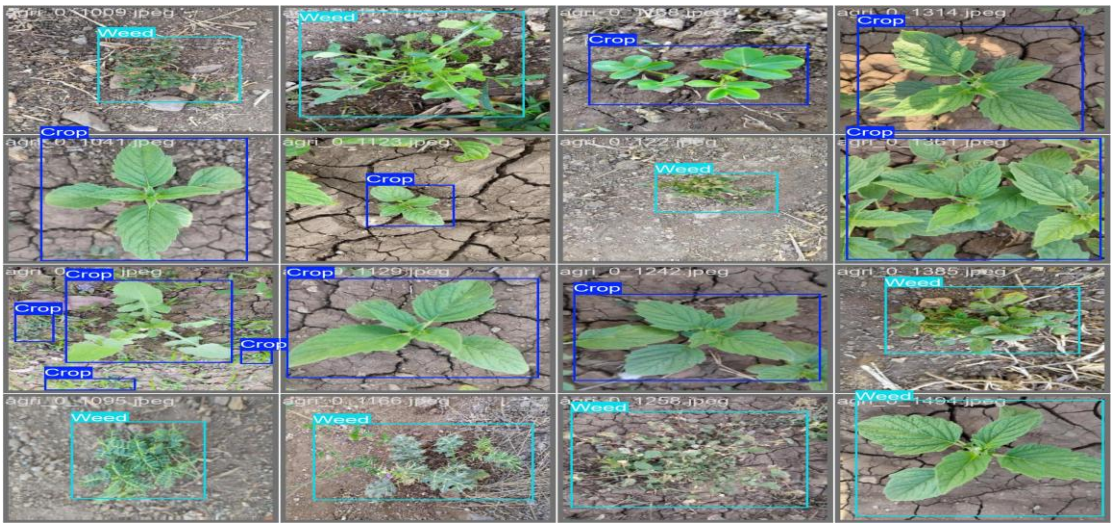

Figure:4.9 Outputs of Weed and Crop Detection with Confidence Scores - 2

Observations:

Detection results also prove the efficiency of the method since it successfully identifies most areas occupied by either weeds or crops, with high confidence scores greater than 0.8. That means that apart from detecting the object, the method demonstrates high confidence when doing so. Besides, detection occurs quite successfully in different field conditions; for example, backgrounds may differ and have a variety of different soils or vegetations which makes it hard to distinguish between different types of plants. Furthermore, even in cases when some overlapping is present in images, detection occurs effectively.

However, a few problems appear in specific cases. First, the method has problems with recognizing extremely small plants as they contain little visual information and therefore cannot be easily distinguished. Second, detection occurs less efficiently when there is poor distinction between weeds and soil. Still, these difficulties do not influence the overall performance of the model and can be considered minor. Thus, the effectiveness of the detection results shows that the method allows precise object localization since the object areas are marked effectively. This means that the detection method works better than typical classification methods.

Confusion Matrix Analysis (Detection Model):

A confusion matrix is a valuable technique that enables users to compare the predicted output classes with the true classes of Crop, Weed, and Background. The confusion matrix is helpful in assessing the quality of the classifier and the degree to which it succeeds in assigning particular categories. With the help of the confusion matrix, one is able to determine the capabilities and drawbacks of the model when used in agricultural scenarios. In general, a confusion matrix will enable one to identify the extent to which the model succeeds in differentiating the classes. The values appearing on the diagonal of the confusion matrix refer to the number of true results. These are the instances where the prediction by the model is successful since it matches the real category. The higher the numbers appearing on the diagonal, the more effective and efficient the model is in distinguishing crops from weeds.

However, the values that are not on the main diagonal indicate the instances where the model wrongly classifies the categories. In such cases, the machine learning model classifies weeds as crops or vice versa because of their similar features. This kind of analysis is essential since it enables us to understand the weaknesses in the machine learning model and improve upon them.

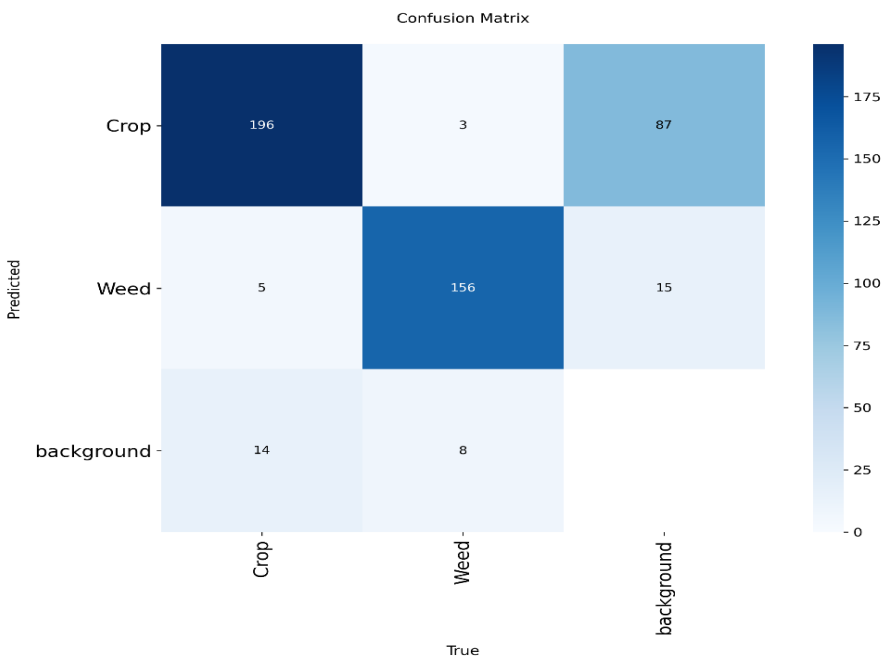

Figure:4.10 YOLOv8 Confusion Matrix

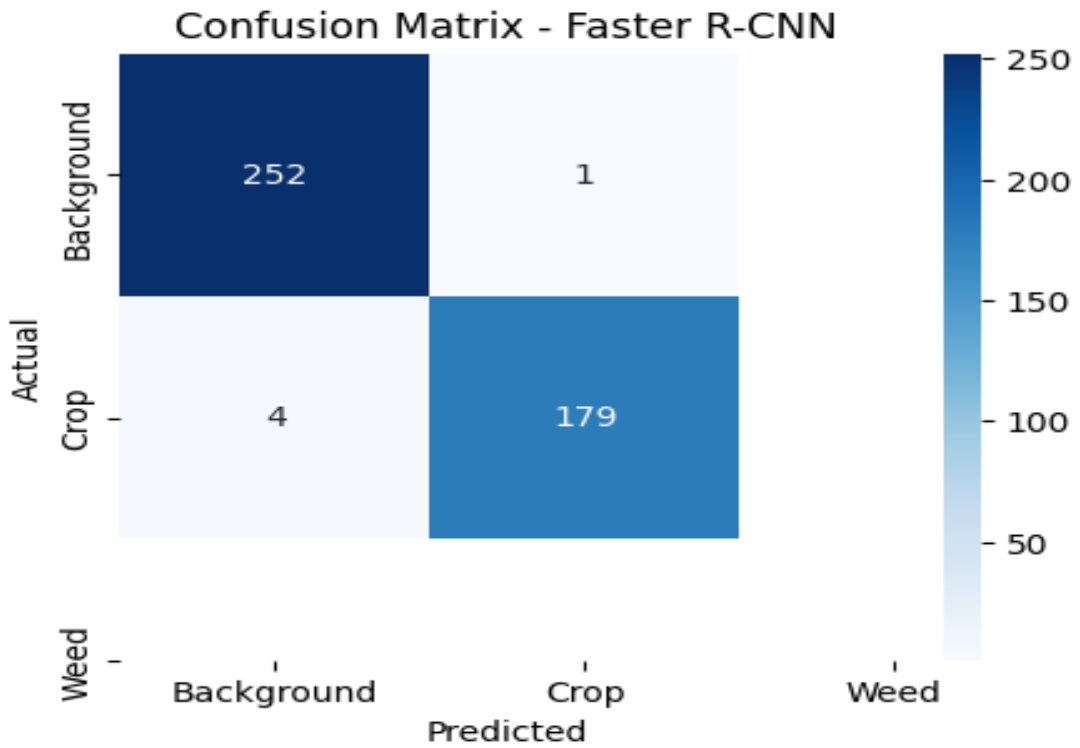

Figure:4.11 Faster R-CNN Confusion Matrix

Observations:

- The confusion matrix is highly dominated by the diagonal, and hence the classification algorithm is able to classify all the data instances with a high level of accuracy.
- All crops and weeds are classified correctly since the machine learning algorithm is capable of identifying these classes.
- False positives and negatives are quite rare for this machine learning algorithm, implying that there are very few erroneous detections by the model.
- There is some confusion with the background category especially at regions that are occupied with the soil due to similarities in appearances.
- Precision and recall are both high, proving the consistency of the results obtained from the detection.

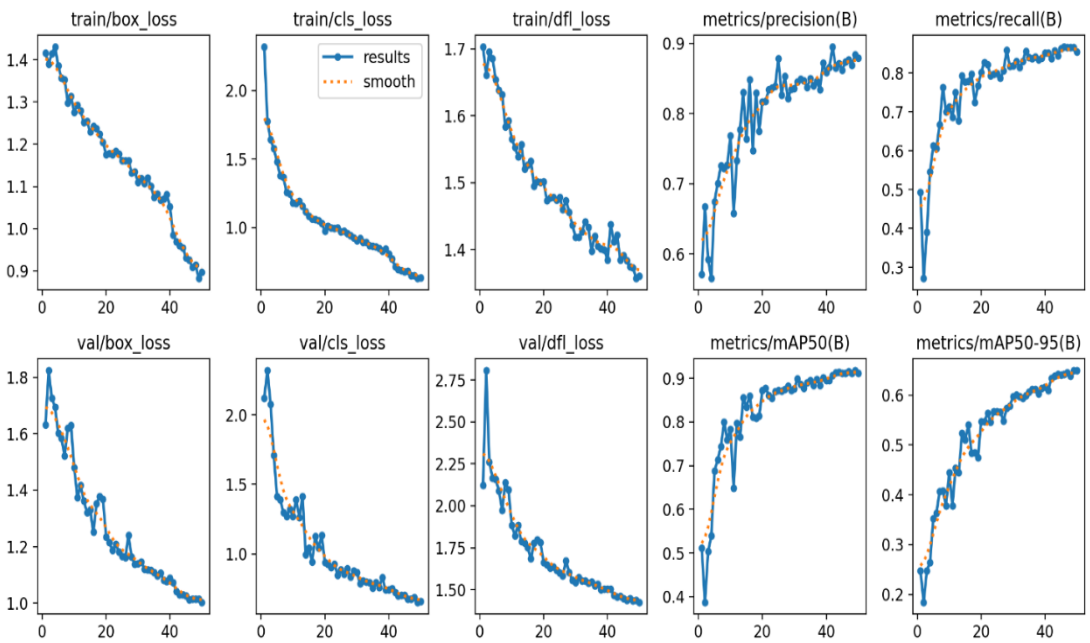

Figure:4.12 Training and Validation Results

Weed Density Estimation:

The measurement of weed density becomes one of the crucial elements of the proposed system due to the fact that apart from identifying weeds, it also measures how severe the problem of weed infestation is. As opposed to classification systems that can only detect whether there are weeds in an image or not, our system provides additional useful information that allows making certain decisions regarding weed elimination in real life. It should be noted that the estimation of weed density can be made with the help of object detection algorithms such as YOLOv8, Faster R-CNN, and EfficientDet. All crops and weeds within the image are localized, so we can make the necessary calculations to find out what part of the image consists of weeds.

The density of weeds is calculated using the following equation:

$$\text{Weed Density (\%)} = \left(\frac{\text{Number of Weeds Detected}}{\text{Total Objects Detected}}\right) \times 100 \qquad \text{-------- (Eq-1)}$$

In other words, we calculate what percentage of all objects we see in the image represents weeds.

Stepwise Process of Weed Density Estimation:

Step 1: Object Detection

This process begins with the identification of all the objects in the agricultural picture through a deep learning-based detector. It takes the agricultural image as input and draws boundary boxes on each detected object. Each boundary box comes with a label, such as 'weed' and 'crop,' and its corresponding probability.

Step 2: Identification and Classification

After identification, each boundary box is classified as belonging to either the weed or crop class depending on their features, like the shape, texture, and structure of leaves. Classification is essential since density estimation requires the proper detection of weed types among other objects detected.

Step 3: Counting Weed Instances

Finally, the number of identified weeds is estimated by counting the total number of boundary boxes that have the label 'weed.' This means that the number of boundary boxes with the 'weed' label represents the number of weeds in the agricultural image. If there are 5 weed regions detected, for instance, the weed count is 5.

Step 4: Counting Total Objects

At this stage, all objects detected in the image are counted. They include the crops and the weeds. The sum of the objects gives the total vegetation in the image. For instance, if there are 5 weeds and 10 crops detected in an image, the total number of objects would be 15.

Step 5: Applying Density Formula

To find the density of the weeds, we divide the number of weeds detected by the total number of objects and multiply by 100%. This formula will change the value to a percentage.

Example:

Weed Density = (5/15) × 100 = 33.3%

Step 6: Density Classification

In addition, the computed percentage density of weeds is grouped into predetermined categories to facilitate interpretation. This helps understand the degree of weed infestation in various parts of the field. This way, proper decisions can be made regarding weed management.

Density Levels and Recommendations:

Low Density (0–30%)

In this case, there are very few weeds relative to the number of crops. Weed influence on the growth of crops is minimal, and no action needs to be taken immediately. Natural growth of crops alone might help limit weed spread.

Medium Density (30–60%)

The appearance of this percentage means that weeds are significantly present and may begin influencing the yield of the crops. Weeding or removal of these weeds should be performed manually in order to avoid their further spread and save effort later on.

High Density (60–100%)

High occurrence of weeds signifies serious weed infestation. At this level, they begin competing with crops strongly for food sources. Chemical means like herbicides spraying should be used to remove the weeds in large quantities efficiently and quickly.

4.2 Result analysis and Test case results

This part covers the detailed performance evaluation of deep learning models used in Weed vs Crop Classification System. The two models that have been considered for evaluation are ResNet50 and Dinov2; both of which have been fine-tuned with the help of a dataset which comprised balanced sets of images of different types of weeds and crops. Different lighting conditions, soil background, leaf shapes, and plant development stages have been considered in the dataset.

Performance Evaluation Metrics

The following performance measures were applied to evaluate the classification accuracy and generalization ability of the models:

Accuracy Score

This statistic calculates the number of correct classifications among the total number of testing samples. The higher the accuracy value, the better the classification model can classify weed and crop images.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \qquad \text{-------- (Eq-2)}$$

Precision

Precision refers to the percentage of images classified under a particular category that are correctly identified. The higher the precision value, the less error there is in misclassifying images as belonging to a different category.

$$\text{Precision} = \frac{TP}{TP+FP} \qquad \text{-------- (Eq-3)}$$

Recall (Sensitivity)

Recall is calculated by taking the ratio of the total number of images for a category correctly classified by the model to the total number of actual instances of the category. The higher the value, the less the number of true instances left unidentified.

$$\text{Recall} = \frac{TP}{TP+FN} \qquad \text{-------- (Eq-4)}$$

F1-Score

The F1 score is a combination of precision and recall. It is particularly important when dealing with imbalanced datasets.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \qquad \text{-------- (Eq-5)}$$

ResNet50 Results

There is an evident enhancement in the efficiency of the model during its training period. There is a considerable increase in the accuracy of the training dataset from 60%

to 93%. Meanwhile, there is also a considerable enhancement in the accuracy of the validation dataset from 61% to 91%. This implies that the model is capable of performing well on datasets that are yet to be seen. It can also be noted that both the loss values for the training and validation datasets decline and remain close to each other. In addition to that, the diagonal dominance of the confusion matrix shows that the weed and crops images are correctly identified by the model.

Dinov2 Results

Furthermore, it is observed that the DINOv2 model can provide consistent and reliable results during the training and evaluation process. During training, the training accuracy ranges between 60% and 89%, and the validation accuracy rises from 58% to 85%, showing effective generalization capability. In addition, the loss graphs are found to reduce rapidly during the early stages, and then, they tend to stabilize within 0.28 and 0.34, showing rapid and stable convergence. In addition, the least difference between the training and validation graphs indicates that the model follows a balanced approach to training without suffering from overfitting or underfitting issues. Besides, the confusion matrix shows the presence of a negligible number of false positives and false negatives, which shows that the model can effectively differentiate visually similar plants.

Accuracy in predictions and reliability in recognizing weeds and crops were high both in ResNet50 and Dinov2 architectures.

- Higher overall accuracy was reached by ResNet50; hence, this algorithm may be used in classification processes.
- High robustness in difficult conditions was guaranteed by Dinov2 due to the attention mechanism.

As seen from the results of the metrics, the confusion matrix, and the tests conducted, the system can be applied in the field.

The project uses sophisticated image recognition algorithms based on advanced deep learning – ResNet50 and DiNoV2 – to help in distinguishing between the weed and crop plants in the agricultural field. Using quality images taken by camera of plants' leaves and entire field, the models use visual cues like texture, color distribution, and

shape of leaves to perform the classification process automatically. Amongst several models used in this process, the model ResNet50 performed the best with the accuracy score of 89% while DiNoV2 model secured 87% accuracy, indicating the robustness of these models for use in farms. The high scores obtained in terms of precision, recall, and f-score in evaluation process suggest that the system can effectively identify most of the weed and crops. This automation would help farmers reduce effort, prevent wastage of herbicides and increase productivity.

In this section, we evaluate in detail the performance of the deep learning models in the Weed and Crop Detection with Density Estimation system. In this study, YOLOv8, Faster R-CNN, and EfficientDet are some of the models that are being used. These models have been trained using a set of annotated agricultural images having bounding boxes for both the weeds and the crops. The images have been chosen with an aim of incorporating real-life situations like varied light conditions, soil backgrounds, varying plant densities, plant overlap, and various growth stages. This helps in enhancing the performance of the models.

The current system is a much more advanced version than the previous mini project in terms of functionality. While the previous version was limited to mere classification of images, the current one detects objects in addition to estimating weed density.
From the results, it can be noted that the detection ability of these models is quite efficient, with mAP scores of between 85% and 92%, indicating high accuracy. Also, the high precision rates reveal that most of the detected weeds and crops are actually correctly classified, as there is a very low percentage of false positive classifications. Furthermore, the high recall rates ensure that almost all weeds and crops found in the images are detected.

It can be noted from the training process that the loss rates are declining at a good pace, thus confirming that these models have been trained in the right manner. One other thing that becomes evident when using these models is that they work efficiently even under complex field conditions such as those involving overlapping plants, different soil textures, and diverse lighting.

Detection Output Analysis

- The algorithm is able to detect several types of weed and crop from a single image.
- The detections come with label and confidence score for better accuracy.
- The system operates efficiently even in unfavorable conditions like low contrast and thick vegetations.

The weed detection models of the proposed system exhibit robustness and consistency in terms of identifying and locating weeds. By exhibiting high values in terms of mAP, precision, and recall, the models prove to be efficient in terms of object detection and minimization of errors. Another strength of this system lies in its ability to provide estimation of the weed density that makes the results more relevant and informative about the extent of weed contamination in the given area.

When comparing the mini project carried out earlier, which dealt only with weed classification, it can be seen that the present system is much more sophisticated and practical for real-world applications. The main difference between the two models is in the fact that the former one provided classification results while the latter not only classifies the objects but also offers suggestions regarding what needs to be done depending on the level of weed density.

4.3 Observations from the work

The entire research into the two projects (mini project involving weed-crop classification and the major project dealing with weed detection with density estimation) shows just how powerful deep learning techniques can be when it comes to solving practical agricultural challenges. In the case of the mini project, the models managed to capture the distinction between the weed and crop images during training, which is evident in the smooth and stable nature of the training procedure. ResNet50 was able to score higher accuracy rates, and demonstrated its reliability in the process, as there was hardly any discrepancy between the training and validation accuracies. Just like ResNet50, DINOv2 delivered accurate outputs, which it did without much struggle, even when dealing with changes in illumination, background, and plant anatomy.

In the major project, the system is capable of going beyond classification as it uses object detection technology that can detect and locate the weed and crop instances present in an image. The object detection models were successful in detecting and localizing the plants under difficult circumstances such as varied soil backgrounds, overlapped crops and various sizes of plants. The training process was also stable, as evidenced by declining loss rates during the training phase. An interesting thing about the system is its capability of detecting objects in multiple scales where it can detect both large and small instances of weeds. Minor errors in very dense and low contrast areas were noted, but the performance of the system in detection tasks was impressive.

An important feature in the main project is that of estimating weed density, which makes the whole system an effective decision-support system. The system can calculate the density of weed by dividing the number of weeds identified with the total number of plants and determine whether there are low, moderate, and high infestations of weeds, and generate recommendations on what needs to be done depending on each situation, such as not doing anything, manual weeding, or spraying of herbicides. In summary, the system as a whole has proved to be very reliable and applicable.

CHAPTER-5
CONCLUSION AND FUTURE STUDY

5.1 Conclusion

This project is able to clearly showcase how the deep learning approach is applied to enhance the agricultural sector through automatic weed and crop detection and analysis. During the mini-project stage, various image classification models including ResNet50 and DINOv2 were developed to classify weeds from crops. These models demonstrated very high accuracy and exhibited good learning ability by detecting visual features such as structure, texture, and colors of the leaves of crops and weeds. It is evident that consistency in their performance during the training and validation phases ensured that they had the generalization ability to correctly classify weeds and crops using new data.

In the major project stage, the current system was improved by developing object detection models, such as YOLOv8, Faster R-CNN, and EfficientDet. Contrary to classification models, detection models can detect multiple objects in an image using bounding boxes. In addition, the models are able to recognize and locate weeds and crops regardless of the number of plants appearing in an image. These improvements make the system more applicable in actual agricultural sites where there are multiple crops and weeds in one image. Besides, object detection models performed very accurately under challenging conditions such as light and dark soils.

Among other major accomplishments of this project is the introduction of weed density assessment that makes it possible to introduce the concept of smart analysis into the design. Through the ratio of weed numbers over plant numbers, the device provides a quantitative assessment of the density of weed population that is categorized into low, medium, and high levels. Based on that, specific suggestions, including no measures taken, hand-picking, or chemical treatment, can be made. With the introduction of weed density estimation, the system becomes an advanced decision-making tool allowing farmers to undertake appropriate actions regarding the issue efficiently.

In conclusion, it should be noted that the offered system provides a comprehensive solution to the problem of weed infestations on farms. It significantly simplifies the process, increases its precision and allows making decisions based on the information received. Moving from the classification stage to detection and density estimation marks a crucial advance that enhances the efficiency of the process. In terms of future development, real-time implementation using drones or mobile apps, integration with IoT networks, and region-wise analysis are possible directions to develop this work further.

5.2 Future Study

The presented approach to automatic detection and estimation of density of weeds provides good grounds for developing intelligent systems for agriculture; however, there are multiple opportunities to improve the existing framework and make it even more efficient. The first obvious area for future research involves implementation of the developed model in a combination with technologies for gathering real-time information from agricultural land, such as UAVs or drone-based systems. The application of drones for obtaining live images from the field will allow performing real-time analysis and making necessary decisions.

Another interesting direction to pursue in the process of further developing the model lies in increasing its accuracy when analyzing photos of weeds grown in extremely complicated environments. Some of the most advanced algorithms, such as the technique of instance segmentation, can be considered for the purpose. Furthermore, an increase in dataset sizes, with greater diversity regarding crops, types of weeds, weather conditions, and seasons, may be useful for achieving higher precision.

Another improvement to the system is through the inclusion of regional weed density estimation and mapping, wherein the farmland is partitioned into smaller regions with separate density calculations. This approach will aid in effective weed removal while minimizing unnecessary spraying of herbicides, hence advancing the concept of precision agriculture. Inclusion of a geographic information system will enhance

visualization of the weed distributions in large farms using a map that will enable the farmers to utilize their resources effectively.

In the future, development of web/mobile applications for the system may prove helpful to the farmers. Such an application will be user friendly with a good interface and will enable the uploading of pictures and receive instant feedback on their status, including advice on how to manage the weeds. Integration of such a system with automatic weed spray equipment and robots will aid in efficient removal of weeds in large farmlands.

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APPENDIX

```
yolo_model = YOLO("yolov8n.pt")
yolo_model.train(
    data=DATA_YAML,
    epochs=30,
    imgsz=640,
    batch=16
)
y_true = []
y_pred = []
results = yolo_model.predict(TEST_IMG)
for r in results:
    img_name = os.path.basename(r.path)
    label_file = os.path.join(TEST_LABEL, img_name.replace(".jpg",".txt"))
    preds = r.boxes.cls.cpu().numpy().astype(int)
    true = []
    if os.path.exists(label_file):
        with open(label_file) as f:
            for line in f:
                true.append(int(line.split()[0]))
    min_len = min(len(true), len(preds))
    y_true.extend(true[:min_len])
    y_pred.extend(preds[:min_len])
import torchvision
from torch.utils.data import Dataset, DataLoader
class WeedDataset(Dataset):
    def __init__(self, img_dir, label_dir):
        self.img_dir = img_dir
        self.label_dir = label_dir
```

```
        self.images = os.listdir(img_dir)
    def __len__(self):
        return len(self.images)
    def __getitem__(self, idx):
        img_name = self.images[idx]
        img_path = os.path.join(self.img_dir, img_name)
        label_path = os.path.join(self.label_dir, img_name.replace(".jpg",".txt"))
        img = cv2.imread(img_path)
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        h, w, _ = img.shape
        img = torch.tensor(img/255., dtype=torch.float32).permute(2,0,1)
        boxes = []
        labels = []
        if os.path.exists(label_path):
            with open(label_path) as f:
                for line in f:
                    cls, x, y, bw, bh = map(float, line.split())
                    x1 = (x - bw/2) * w
                    y1 = (y - bh/2) * h
                    x2 = (x + bw/2) * w
                    y2 = (y + bh/2) * h
                    boxes.append([x1,y1,x2,y2])
                    labels.append(int(cls)+1)
        target = {
            "boxes": torch.tensor(boxes, dtype=torch.float32),
            "labels": torch.tensor(labels, dtype=torch.int64)
        }
        return img, target
train_dataset = WeedDataset(
    "/content/dataset/images/train",
```

```
    "/content/dataset/labels/train"
)
train_loader = DataLoader(
    train_dataset,
    batch_size=4,
    shuffle=True,
    collate_fn=lambda x: tuple(zip(*x))
)
model = torchvision.models.detection.fasterrcnn_resnet50_fpn(pretrained=True)
num_classes = 3
in_features = model.roi_heads.box_predictor.cls_score.in_features
model.roi_heads.box_predictor
=torchvision.models.detection.faster_rcnn.FastRCNNPredictor(
    in_features, num_classes
)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
optimizer = torch.optim.Adam(model.parameters(), lr=0.0001)
for epoch in range(5):
    model.train()
    total_loss = 0
    for images, targets in train_loader:
        images = [img.to(device) for img in images]
        targets = [{k:v.to(device) for k,v in t.items()} for t in targets]
        loss_dict = model(images, targets)
        loss = sum(loss_dict.values())
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()
        total_loss += loss.item()
```

```
    print(f"Epoch {epoch+1}, Loss: {total_loss}")
y_true = []
y_pred = []
for img_name in os.listdir(TEST_IMG):
    img_path = os.path.join(TEST_IMG, img_name)
    label_file = os.path.join(TEST_LABEL, img_name.replace(".jpg",".txt"))
    img = cv2.imread(img_path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    img_tensor = torch.tensor(img/255., dtype=torch.float32).permute(2,0,1).to(device)
    with torch.no_grad():
        output = model([img_tensor])
    preds = output[0]['labels'].cpu().numpy()
    true = []
    if os.path.exists(label_file):
        with open(label_file) as f:
            for line in f:
                true.append(int(line.split()[0]) + 1)
    min_len = min(len(true), len(preds))
    y_true.extend(true[:min_len])
    y_pred.extend(preds[:min_len])
```

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The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

17% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text

Match Groups

194 Not Cited or Quoted 17%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

10% Internet sources
8% Publications
14% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.